cp_10a

1

Multiplication Table | 30/30 Solved

Write a program to print the multiplication table of the given number (N) up to ten multiples in the format "N x i = M".

Input

The first line of input will contain an integer.

Output

The output should be ten lines containing the multiples in the given format.

code:

```
num=int(input())
str_num=str(num)
for i in range(1,11):
    prod=num*i
    i_str=str(i)
    str_prod=str(prod)
    table=str_num+" x "+i_str+" = "+str_prod
    print(table)
```

2

Vowels in a String | 30/30 Solved

Easy

Given a string `S` , write a program to find the vowels in the given string `S` .
Print the resultant string by joining all the vowels in the string `S` .

Note

The Vowels are **a**, **e**, **i**, **o**, and **u**.

Input

The input will be a single line containing a string representing `S` .

Output

The output should be a single line containing a string obtained by joining the vowels in the string `S` .

Explanation

For example, if the given string is `S = "container"` ,

- The characters in the string **container** are **c**, **o**, **n**, **t**, **a**, **i**, **n**, **e** and **r**.
- The vowels in the string **container** are **o**, **a**, **i** and **e**.
- The string obtained by joining all the vowels (**o**, **a**, **i** and **e**) is **oaie**.

The output should be **oaie**.

code:

```
st=input()
s=""
for each_char in st:
    if each_char in("a","e","i","o","u"):
```

```
        s=s+each_char
print(s)
```

3

Reverse the String | 35/35 | Solved

Write a program to print the reverse of the given string.

Input

The input will be a single line containing a string.

Output

The output should be a single line containing the reverse of the given string.

Explanation

For example, if the given string is "Hurray! We have won the match.", your code shoul print the reverse of the string, which is ".hctam eht now evah eW !yarruH".

Sample Input 1

```
Hurray! We have won the match.
```

Sample Output 1

```
.hctam eht now evah eW !yarruH
```

code:

```
str=input()
len_str=len(str)
string=""
for i in range(len_str):
    string=string+str[len_str-1]
```

```
    len_str=len_str-1
print(string)
```

4

Sum of K Powers - 2 | 30/30 Solved

Easy

Given a number `N` , write a program to print the sum of the **Kth** power of all the digits in the given number.

K indicates the number of digits of the number `N` .

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing an integer that is the sum of the **Kth** power of all the digits of the number `N` .

Explanation

For example, if the given number is `N = 24753` ,

- The digits of **24753** are **2**, **4**, **7**, **5**, and **3**.
- The number of digits of **24753** is **5**. So, the value of **K** is **5**.
- The **5** (**K**) power of all the digits of **24753** are **32**, **1024**, **16807**, **3125**, and **243**.
- The sum of the **5** (**K**) power of all the digits is **21231**. ($2^5 + 4^5 + 7^5 + 5^5 + 3^5 =$ **21231**)

The output should be **21231**.

code:

```python
num=int(input())
num_str=str(num)
num_len=len(num_str)
sum=0
for i in range(num_len):
    i_val=int(num_str[i])
    sum=sum+i_val**num_len
print(sum)
```

5

Armstrong Number - 3 | 30/30

Solved

Write a program that reads a number `N` and checks if `N` is an Armstrong Number.

Print **Armstrong Number** is the given number is an Armstrong Number. Otherwise, print **Not an Armstrong Number**.

Note

Armstrong Number

A number is said to be an **Armstrong Number** if the sum of powers of all the digits of the number is the number itself.

Example:

1634 is an **Armstrong Number** as the sum of powers of all digits of `N` is equal to **1634**. ($1^4 + 6^4 + 3^4 + 4^4 = 1634$)

Input

The input will be a single line containing an integer.

Output

The output should be a single line containing a string. **Armstrong Number** should be printed if the given number is an Armstrong Number. Otherwise, print **Not an Armstrong Number**.

Explanation

For example, if the given number is **54748**,

- The digits in **54748** are **5**, **4**, **7**, **4** and **8**.
- Number of digits in **54748** is **5**.
- Sum of digits to the power of **5** is **54748**. ($5^5+4^5+7^5+4^5+8^5$)
- Sum of digits to the power of **5** (**54748**) is equal to the number (**54748**).

The output should be **Armstrong Number**.

code:

```python
num=int(input())
num_str=str(num)
length=len(num_str)
sum=0
for i in range(length):
    each_char=int(num_str[i])
    sum=sum+each_char**length
if sum==num:
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

6

Factors of a Number | 30/30

Easy Hints

Given a number `N` , write a program to print the factors of the number `N` .

Note

Factors of N:

If a number `N` is divisible by `X` , then `X` is a factor of `N` .

Example:

If `N = 15` and `X = 3` , then **3** is a factor of **15** as **15** is divisible by **3**.

If `N = 15` and `X = 7` , then **7** is **not** a factor of **15** as **15** is not divisible by **7**.

Input

The input will be a single line containing a positive integer.

Output

The output should contain all the factors of the given number, each on a new line.

Explanation

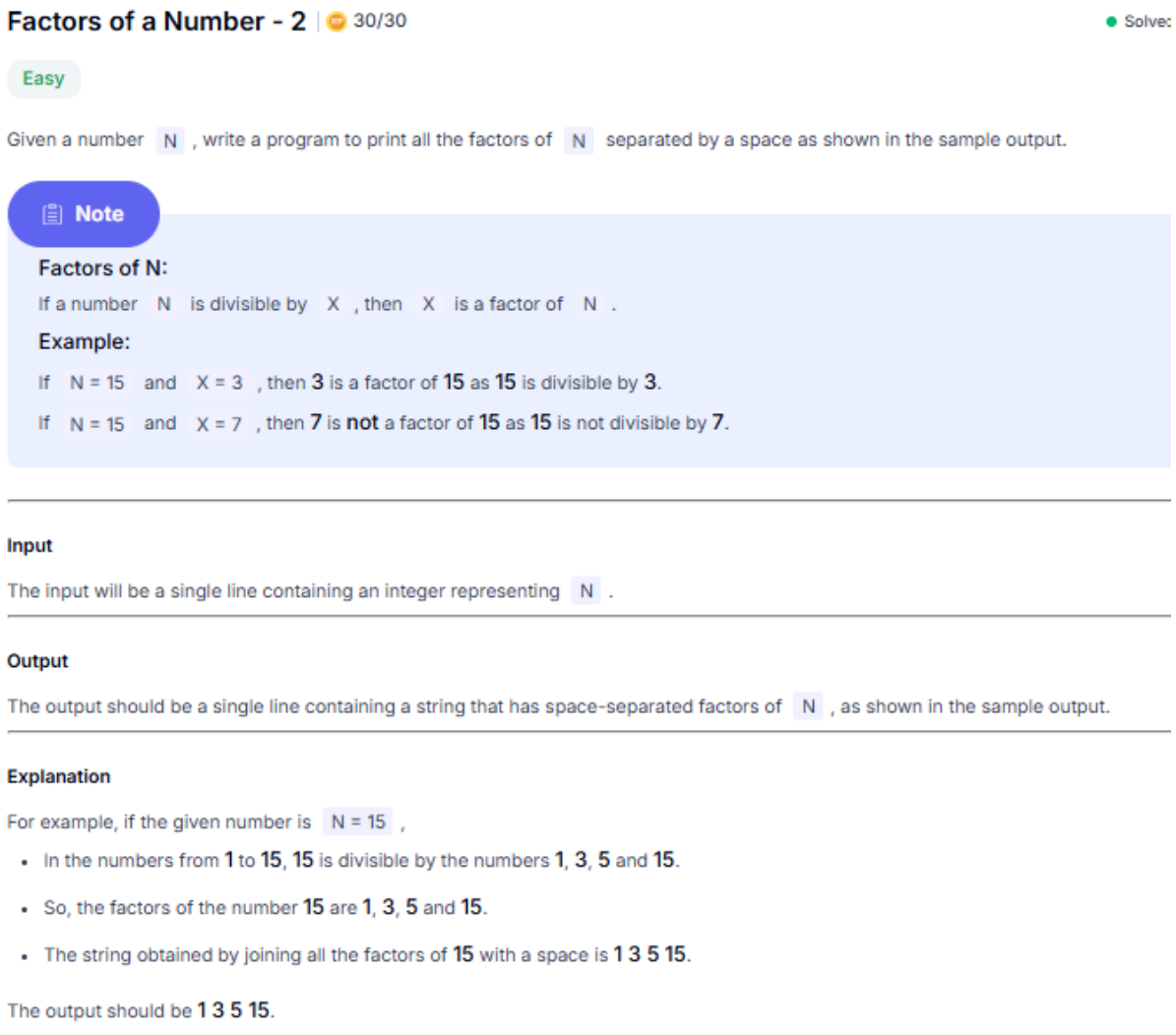
For example, if the given number is `N = 6` ,

- The number **6** is divisible by the numbers **1**, **2**, **3** and **6**.
- So, the factors of the number **6** are **1**, **2**, **3** and **6**.

code:

```
num=int(input())
for i in range(1,num+1):
    factor=num%i
    if factor==0:
        print(i)
```

7



Factors of a Number - 2 | 30/30 ● Solved

Easy

Given a number `N` , write a program to print all the factors of `N` separated by a space as shown in the sample output.

Note

Factors of N:

If a number `N` is divisible by `X` , then `X` is a factor of `N` .

Example:

If `N = 15` and `X = 3` , then **3** is a factor of **15** as **15** is divisible by **3**.

If `N = 15` and `X = 7` , then **7** is **not** a factor of **15** as **15** is not divisible by **7**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing a string that has space-separated factors of `N` , as shown in the sample output.

Explanation

For example, if the given number is `N = 15` ,

- In the numbers from **1** to **15**, **15** is divisible by the numbers **1**, **3**, **5** and **15**.
- So, the factors of the number **15** are **1**, **3**, **5** and **15**.
- The string obtained by joining all the factors of **15** with a space is **1 3 5 15**.

The output should be **1 3 5 15**.

code:

```
num=int(input())
str_num=str(num)
```

```
factors=""
for i in range(1,num+1):
    factor=num%i
    if factor==0:
        factors=factors+str(i)+" "
print(factors)
```

8

Sum of All Factors | 30/30

Given a number `N` , write a program to print the sum of all the factors of the number `N` .

Note

Factors of N:

If a number `N` is divisible by `X` , then `X` is a factor of `N` .

Example:

If `N = 15` and `X = 3` , then **3** is a factor of **15** as **15** is divisible by **3**.

If `N = 15` and `X = 7` , then **7** is **not** a factor of **15** as **15** is not divisible by **7**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing an integer that is the sum of the factors of `N` .

Explanation

For example, if the given number is `N = 12` ,

- The number **12** is divisible by the numbers **1**, **2**, **3**, **4**, **6**, and **12**.
- So, the factors of the number **12** are **1**, **2**, **3**, **4**, **6**, and **12**.
- The sum of all the factors of **12** is **28** (**1** + **2** + **3** + **4** + **6** + **12**).

The output should be **28**.

code:

```
num=int(input())
sum=0
for i in range(1,num+1):
```

```
    factor=num%i
    if factor==0:
        sum=sum+i
print(sum)
```

9

Greatest Among N Numbers | 40/40

Given a number `N` , write a program that reads `N` inputs and prints the greatest number among the given inputs.

Input

The first line of input contains an integer representing `N` .
The next `N` lines of input contain integers.

Output

The output should be a single line containing an integer that is the greatest among the given numbers.

Explanation

For example, if the given number is `N = 5` ,

- The **5** inputs are **8**, **11**, **96**, **49**, and **25**.
- The maximum of the input **8** is **8**.
- The maximum of the inputs **8** and **11** is **11**.
- The maximum of the inputs **8**, **11** and **96** is **96**.
- The maximum of the inputs **8**, **11**, **96** and **49** is **96**.
- The maximum of the inputs **8**, **11**, **96**, **49** and **25** is **96**.
- The greatest among the given numbers is **96**.

The output should be **96**.

code:

```
num=int(input())
first_num=int(input())
greatest=first_num
for i in range(num-1):
    nums=int(input())
    if nums>greatest:
```

```
        greatest=nums
print(greatest)
```

10

Perfect Number | 30/30 — Solved

Write a program to check whether the given number is a perfect number or not.
A number is considered as a Perfect number if sum of all factors excluding itself is equal to the number.

Input

The input will be a single line containing a positive integer.

Output

The output should be a single line containing either "Perfect Number" or "Not a Perfect Number".

Explanation

For example, if the given number is 6, the factors of 6 excluding it are 1, 2, and 3. The sum of these factors is 6, which is equal to the giv number. So the output should be "Perfect Number".

Similarly, if the given number is 21, the factors of 21 excluding it are 1, 3, and 7. The sum of these factors is 11, Which is not equal to the given number. So the output should be "Not a Perfect Number".

Sample Input 1

```
6
```

Sample Output 1

```
Perfect Number
```

code:

```
num=int(input())
sum=0
for i in range(1,num):
    factors_num=num%i
    if factors_num==0:
        sum=sum+i
if sum==num:
    print("Perfect Number")
else:
    print("Not a Perfect Number")
```